Project Scope: Company-Specific RAG Chatbot Powered by Fine-Tuned LLM

Project Name

GaruanCDX Internal & Client-Facing Knowledge Assistant

(Codename: GarunaKnow or CDX Insight Bot)

Version: 1.0 – March 2026

Prepared By: Aadarsh Mishra– GaruanCDX leadership & development team

Objective: Deliver a production-grade, secure Retrieval-Augmented Generation (RAG) chatbot that connects to proprietary company data (internal docs, client project knowledge, code repos, policies, etc.) and leverages a **fine-tuned LLM** for domain-adapted reasoning, tone, and accuracy. The system must serve both internal employees (knowledge access, onboarding, support) and selected clients (via the ClearTrack portal extension).

1. Business Context & Strategic Value

GaruanCDX, as a custom software & digital solutions agency, generates massive proprietary knowledge: project specs, code snippets, client requirements, architecture decisions, lessons learned, SOPs, compliance docs, past proposals, and client-specific data.

Current pain points:

- Knowledge is scattered (Google Drive, Notion, Git repos, emails, Jira/ClickUp).
- New hires & developers waste 20–40% of time searching.
- Client queries repeat → support overhead.
- Generic LLMs hallucinate on company-specific terminology, past decisions, or GaruanCDX style.

Value created by this chatbot

- Reduce internal knowledge search time by 50–70% (benchmark from similar enterprise RAG deployments 2025–2026).
- Accelerate client onboarding & support in ClearTrack portal.
- Position GaruanCDX as an AI-forward agency → showcase in proposals & on garuancdx.com.
- Enable future monetization: white-label version for clients or premium add-on.

Success Metrics (KPIs)

- Answer accuracy (grounded in retrieved docs): ≥ 92% (measured via human eval + RAGAS-like metrics)
- Hallucination rate: ≤ 5%
- User satisfaction (CSAT): ≥ 4.2/5
- Average query resolution time: < 15 seconds
- Adoption: ≥ 70% of team using weekly within 3 months
- Cost per query: < ₹0.50–1.50 (target with efficient fine-tuned model)

2. Core Architecture – Hybrid Fine-Tuned LLM + Advanced RAG (2026 Best Practice)

Modern enterprise RAG in 2026 combines:

- **Fine-tuning** for domain tone, terminology, reasoning style, safety (reduce hallucinations on company data).
- **RAG** for dynamic, up-to-date, auditable knowledge (no retraining when docs change).

Recommended Pattern: Retrieval-Augmented Fine-Tuning (RAFT) hybrid

- Fine-tune base model to become better at using retrieved context & following company voice.
- Use RAG for factual grounding → avoids catastrophic forgetting & constant retraining.

High-Level Components

1. **Data Ingestion & Processing Pipeline**
2. **Embedding & Vector Store**
3. **Fine-Tuned LLM Inference**
4. **Retrieval Pipeline (with re-ranking & query transformation)**
5. **Chat Orchestration & Guardrails**
6. **Frontend Interfaces** (internal web + ClearTrack integration)
7. **Observability, Evaluation & Continuous Improvement**

3. Detailed Requirements & Implementation Phases

Phase 1 – Foundation (Weeks 1–4) – MVP Data & Retrieval

Must deliver: Working RAG with off-the-shelf LLM before fine-tuning.

- **Data Sources** (connectors):
  - Google Drive / Shared folders (project docs, proposals)
  - Notion / Confluence pages
  - Git repositories (READMEs, code comments via git clone + parsing)
  - PDFs, DOCX, Markdown, CSV (invoices, requirements)
  - Future: Jira tickets, Slack threads (via export/API)
- **Ingestion Pipeline**
  - Loaders: LlamaIndex / LangChain document loaders
  - Pre-processing: Clean text, remove noise, extract metadata (project ID, client name, date, author)
  - Chunking strategy: Semantic chunking (200–800 tokens) + 20–30% overlap → hierarchical (small + parent chunks)
  - Embeddings: text-embedding-3-large (OpenAI) or bge-m3 / gte-Qwen2-7B-instruct (open, multilingual support for India context)
- **Vector Database** (choose one):

- Pinecone serverless (recommended for scale & hybrid search)
- Weaviate / Qdrant (open-source, good metadata filtering)
- pgvector + Supabase (if already using Supabase stack)

- **Basic Retrieval**: Hybrid (dense + BM25) → Cohere Rerank / bge-reranker-large (top-3 to top-8 chunks)

Deliverable: Internal test chatbot answering factual questions from uploaded docs (no fine-tuning yet).

Phase 2 – Fine-Tuning the LLM (Weeks 5–9)

Goal: Adapt base model to GaruanCDX domain (software dev terminology, client project language, professional yet approachable tone, Maharashtra/Indian English nuances).

Model Selection (2026 sweet spot – cost vs performance):

- Base: Llama-3.1-8B-Instruct or Qwen2.5-7B-Instruct or Mistral-Nemo-Instruct-2407 (strong multilingual)
- Alternative premium: Command R+ or Grok-2-mini (if API budget allows)
- Fine-tuning method: LoRA / QLoRA (efficient, 4-bit/8-bit) via Unsloth / Axolotl / Llama-Factory

Training Data Preparation (critical – 1,000–5,000 high-quality examples)

- Synthetic data: Use strong LLM (GPT-4o / Grok / Claude 3.5) to generate Q&A pairs from your docs.
- Real data: Curate 200–500 real internal questions + ideal answers.
- RAFT-style: Include "distractor" chunks → train model to ignore noise.
- Formats: ChatML / instruction-following + system prompt with company voice.

Fine-Tuning Goals

- Better context usage (less ignoring retrieved docs)
- Consistent tone (professional, helpful, concise)
- Domain vocabulary (React hooks, microservices, offshore dev pricing, etc.)
- Refusal behavior on sensitive/out-of-scope queries

Hosting Inference

- vLLM / TGI (fast, OpenAI-compatible API)
- Deploy: Railway / Modal / RunPod / AWS SageMaker (GPU spot instances)

Phase 3 – Advanced RAG & Orchestration (Weeks 10–14)

- **Query Transformation**: HyDE, query decomposition, step-back prompting
- **Re-ranking & Fusion**: Multi-retriever fusion + Cohere/bge reranker
- **Post-retrieval**: Contextual compression, chunk fusion
- **Agentic layer** (optional stretch): Tool use (e.g., query Jira API for live ticket status)
- **Guardrails**: LlamaGuard / NeMo Guardrails – block PII leaks, harmful content

- **Memory**: Conversation summary + entity memory (LangGraph / LlamaIndex)

Phase 4 – User Interfaces & Security (Weeks 15–18)

- **Internal**: Streamlit / Gradio / Next.js chat UI + SSO (Clerk / Supabase Auth)
- **Client-facing**: Embed in ClearTrack portal (iframe or React component) – role-based access (only see own projects)
- **Security & Compliance**
  - Data encryption at rest/transit
  - RBAC (project-level isolation)
  - Audit logs (every query + retrieved chunks)
  - DPDP Act 2023 compliance (India): consent, data minimization, right to erase

Phase 5 – Evaluation, Monitoring & Iteration (Ongoing)

- **Offline eval**: RAGAS (faithfulness, answer relevance, context precision)
- **Online eval**: Human thumbs-up/down + implicit signals (query abandonment)
- **Observability**: LangSmith / Phoenix / Helicone
- **Continuous improvement**: Feedback → synthetic data → periodic LoRA updates

4. Tech Stack Recommendation (Balanced for Speed, Cost & Control – 2026)

- Orchestration: LlamaIndex (strong enterprise focus) or LangChain/LangGraph
- Embeddings: OpenAI / Voyage AI / open Indian-friendly (bge-m3)
- Vector DB: Pinecone serverless or pgvector
- LLM Fine-tune: Unsloth + Hugging Face Hub
- Inference: vLLM on GPU
- Frontend: React + shadcn/ui (chat UI)
- Auth & Hosting: Supabase / Vercel + Railway/Modal for GPU
- Monitoring: LangSmith / Prometheus + Grafana